



MYTHIC

Model Summary Report

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1 Changelog

- Version 3.0
 - Added BEVFormer 1600x900
 - Model latency and throughput include digital NPU measured as a sequential pipeline. Support for streaming mode (parallel execution of analog and digital NPU) will be added in Phase 3.5.
- Version 2.0
 - Added YOLOv8-IS model metrics
 - Updated accuracy metrics to use Monte-Carlo simulation results
 - Updated latency and throughput metrics based on the latest compiler results
 - Updated power metrics for the latest FuncSim updates
- Version 1.0
 - Initial version generated for SDK 25.11

2 Introduction

This document describes the models available in Mythic's M2000 (Gen 2 architecture) SDK. The SDK contains seven models: ResNet-50, YOLOv8s Object Detection (OD), YOLOv8s Pose Detection (PD), YOLOv8s Instance Segmentation (IS), YOLOPX Panoptic Vision, Pythia-70m, and BEVFormer-tiny. For each model except Pythia-70n, three artifacts are provided that represent intermediate steps of the model conversion process:

- **FP32 ONNX:** This is a trained FP32 artifact with all unsupported operators replaced and the input tensor size set to meet the product requirements. This artifact is the starting point of Mythic's retraining process.
- **Retrained ONNX:** This is an ANA8 retrained artifact. The ONNX graph includes attributes to distinguish digital from analog nodes. It also includes the trained weights. The Accuracy simulation uses this artifact to predict the final model accuracy.
- **Compiler-ready artifact:** This artifact is generated from the retrained ONNX. The original graph has been split into separate analog (labeled `on_chip`) and digital (labeled `off_chip`) subgraphs. The compiler creates firmware for the analog subgraphs and adds them to a new artifact.

These artifacts are provided to accelerate analysis of power, area, performance, and accuracy. For customers who want to follow the complete retraining process, the SDK contains retraining user guides that describe how to recreate the artifact.

Note that model throughput is lower than in previous releases and is gated by the number of v-MP cores used. Throughput can be increased by increasing the number of cores.

3 Model Summary

The directory tree below shows the location of each model artifact within the `mythic_sdk/v26.05.1/` directory with a comment describing its purpose.

```
models/
├── training
│   └── bevformer
│       ├── bevformer-tiny-1600x900-trained.onnx      # BEVFormer ANA8 retrained ONNX
│       ├── bevformer-tiny-1600x900-trained.tar.gz    # BEVFormer Compiler-ready artifact
│       └── bevformer-tiny-fp32-1600x900.onnx         # BEVFormer FP32 ONNX
│   └── huggingface_classifiers
│       └── resnet50_0.8098_opset11_224x224.onnx      # ResNet-50 FP32 ONNX
```

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├── resnet50_imagenet_trained.onnx	# ResNet-50 ANA8 retrained ONNX
├── resnet50_imagenet_trained.tar.gz	# ResNet-50 Compiler-ready artifact
├── pythia	
│ ├── pythia_70m_fp32.onnx	# Pythia-70m FP32 ONNX
│ └── pythia_70m-trained.onnx	# Pythia-70m ANA8 retrained ONNX
├── yolopx	
│ ├── multiclass_yolopx_fp32.onnx	# YOLOPX FP32 ONNX
│ ├── yolopx_trained.onnx	# YOLOPX ANA8 retrained ONNX
│ └── yolopx_trained.tar.gz	# YOLOPX Compiler-ready artifact
└── yolov8	
├── yolov8s-fp32-1920x1920_t125_e432.onnx	# YOLOv8s OD FP32 ONNX
├── yolov8s-pose_1920x1088_trained.onnx	# YOLOv8s-PD ANA8 retrained ONNX
├── yolov8s-pose_1920x1088_trained.tar.gz	# YOLOv8s-PD Compiler-ready artifact
├── yolov8s-pose_relu_fp32_1920_0.61.onnx	# YOLOv8s-PD FP32 ONNX
├── yolov8s-seg_1920x1088_trained.onnx	# YOLOv8s-IS ANA8 retrained ONNX
├── yolov8s-seg_1920x1088_trained.tar.gz	# YOLOv8s-IS Compiler-ready artifact
├── yolov8s-seg_relu_resize_fp32_1920.onnx	# YOLOv8s-IS FP32 ONNX
├── yolov8s_trained.onnx	# YOLOv8s OD ANA8 retrained ONNX
└── yolov8s_trained.tar.gz	# YOLOv8s OD Compiler-ready artifact

4 Monte-Carlo Accuracy Prediction

This report includes Monte Carlo accuracy predictions. This methodology randomly varies chip-specific parameters across a large number of accuracy simulations to estimate both the average accuracy and the 100PPM worst-case accuracy across a population of devices. The two metrics that are described in each table are:

- **ANA8 Retrained Model:** This is the average accuracy from the Monte-Carlo simulation.
- **ANA8 Retrained Model 100PPM:** The minimum accuracy of 99.99% of all devices assuming a Gaussian accuracy distribution.

⚠ The Monte-Carlo simulation is currently for a single operating temperature. Temperature Compensation and its impact on accuracy will be added to a future release.

5 ResNet-50 v1.5

ResNet-50 uses the original graph as-is. The model is structured as follows:

- **Input Resolution:** 224x224x3
- **Digital Layers:** Final GEMM (Gemm_174). Note that these digital layers are not included in the power and latency estimates shown below. This functionality will be added when the digital NPU/CPU subsystem is defined.

The training parameters for this model are defined below. Note that the Hyperparameters are defined in an SDK configuration file.

- **GPU:** Four Nvidia A6000
- **Dataset:** ImageNet 2012

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- **Hyperparameters:** `$MYTHIC_SDK_ROOT/mythic-model-zoo/configs/huggingface_classifiers/training/training_config/resnet50_imagenet.yaml`
- **Training Guide:** `'doc/user-guides/ResNet-50 Retraining Guide.pdf'`

The table below shows the Top 1 accuracy metrics for the original FP32, ANA8 Quantized Model, and the ANA8 Retrained Model.

Accuracy Metric	FP32 Model	ANA8 Quantized Model	ANA8 Retrained Model	ANA8 Retrained Model 100PPM
Top 1 Prediction	0.81	0.001	0.7722	0.7685

The table below shows the performance, power statistics, and tile statistics for the ANA8 model. For more details on performance, power, and tile statistics (e.g. ACE utilization, Weight utilization, and SRAM utilization per tile), see the document Compiler Optimization Report - ResNet-50.

Attribute	Value
Mythic NPU Configuration	m2024 (24 ACEs, 6 ACE tiles, 4 MB per ACE tile, 127 mm^2 die area) 1 Digital NPU Core NPU/CPU subsystems options are still under consideration
Latency (High Optimization)	0.55 ms (some variance in results is observed)
Frames per second (High Optimization)	1830 fps
Power Consumption (High Optimization)	1.1 W at maximum frame rate (1830 fps)
ACE (compute) utilization	71.6%
Weight (flash cell) utilization	26.8M weights (89.7% of capacity)
SRAM utilization	1.6 MB (6.4% of simulated SRAM capacity)

6 YOLOv8s Object Detection

YOLOv8s has been modified to replace the SiLU activation. The model is structured as follows:

- **Input Resolution:** 1920x1088x3
- **Digital Layers:** DFL head: `Softmax`, `1x16x1x1 Convolution`, `Sigmoid`. Note that these digital layers are not included in the power and latency estimates shown below. This functionality will be added when the digital NPU/CPU subsystem is defined.

The training parameters for this model are defined below. Note that the Hyperparameters are defined in an SDK configuration file.

- **GPU:** Eight Nvidia A6000
- **Dataset:** COCO 2017
- **Hyperparameters:**
 - `$MYTHIC_SDK_ROOT/mythic-model-zoo/configs/yolov8/training/training_config/yolov8s-m2000.yaml` and the files referenced from it.
 - Batch size: $4 \times 7 = 28$
 - For 7 GPUs the required command line parameters are `device='0,1,2,3,4,5,6'`
`training_config.batch=28`
- **Training Guide:** `'doc/user-guide/YOLO Retraining Guide for Mythic AMP.pdf'`

Accuracy Metric	FP32 Model	ANA8 Quantized Model	ANA8 Retrained Model	ANA8 Retrained Model 100PPM
mAP50:95	0.449	0.00	0.429	0.425

The table below shows the performance, power statistics, and tile statistics for the ANA8 model. For more details on performance, power, and tile statistics (e.g. ACE utilization, Weight utilization, and SRAM utilization per tile), see the document Compiler Optimization Report - YOLOv8s Object Detection.

Attribute	Value
Mythic NPU Configuration	m2024 (24 ACEs, 6 ACE tiles, 4 MB per ACE tile, 127 mm ² die area) 4 Digital NPU Cores NPU/CPU subsystems options are still under consideration
Latency (High Optimization)	17.26 ms (some variance in results is observed)
Frames per second (High Optimization)	57 fps

Attribute	Value
Power Consumption (High Optimization)	0.6 W at maximum frame rate (57 fps) 0.3 W at target frame rate (30 fps)
ACE (compute) utilization	75.0%
Weight (flash cell) utilization	18.6M weights (62.3% of capacity)
SRAM utilization	5.2 MB (20.5% of simulated SRAM capacity)

7 YOLOv8s Instance Segmentation

YOLOv8s Segmentation has been modified to replace the SiLU activation. The model is structured as follows:

- **Input Resolution:** 1920x1088x3
- **Digital Layers:** 1x16x1x1 Convolution, 3 32x32x1x1 Convolutions, 32x128x1x1 Convolution, Softmax, Sigmoid. Note that these digital layers are not included in the power and latency estimates shown below. This functionality will be added when the digital NPU/CPU subsystem is defined.

The training parameters for this model are defined below. Note that the Hyperparameters are defined in an SDK configuration file.

- **GPU:** Eight Nvidia A6000
- **Dataset:** COCO 2017
- **Hyperparameters:**
 - \$MYTHIC_SDK_ROOT/mythic-model-zoo/configs/yolov8/training/training_config/yolov8s-seg-m2000.yaml and the files referenced from it.
 - Batch size: 2 x 8 = 16
 - For 8 GPUs the required command line parameters are device='0,1,2,3,4,5,6,7' training_config.batch=16
- **Training Guide:** 'doc/user-guide/YOLO Retraining Guide for Mythic AMP.pdf'

Accuracy Metric	FP32 Model	ANA8 Quantized Model	ANA8 Retrained Model	ANA8 Retrained Model 100PPM
mAP50:95	0.379	0	0.369	0.364

The table below shows the performance, power statistics, and tile statistics for the ANA8 model. For more details on performance, power, and tile statistics (e.g. ACE utilization, Weight utilization, and SRAM utilization per tile), see the document Compiler Optimization Report - YOLOv8s Instance Segmentation.

Attribute	Value
Mythic NPU Configuration	m2024 (24 ACEs, 6 ACE tiles, 4 MB per ACE tile, 127 mm ² die area) NPU/CPU subsystems options are still under consideration
Latency (High Optimization)	8.40 ms (some variance in results is observed)
Frames per second (High Optimization)	119 fps
Power Consumption (High Optimization)	1.3 W at maximum frame rate (119 fps) 0.4 W at target frame rate (30 fps)
ACE (compute) utilization	73.5%
Weight (flash cell) utilization	18.9M weights (63.3% of capacity)
SRAM utilization	5.2 MB (20.5% of simulated SRAM capacity)

8 YOLOv8s Pose Estimation

YOLOv8s-Pose has been modified to replace the SiLU activation. The model is structured as follows:

- **Input Resolution:** 1920x1088x3
- **Digital Layers:** The last 2 layers of the pose keypoints head for each 3 scales (6 layers total), DFL head: `Softmax`, `1x16x1x1 Convolution`, `Sigmoid`. Note that these digital layers are not included in the power and latency estimates shown below. This functionality will be added when the digital NPU/CPU subsystem is defined.

The training parameters for this model are defined below. Note that the Hyperparameters are defined in an SDK configuration file.

- **GPU:** Eight Nvidia A6000
- **Dataset:** COCO 2017 with Pose Annotations
- **Hyperparameters:**
 - `$MYTHIC_SDK_ROOT/mythic-model-zoo/configs/yolov8/training/training_config/yolov8s-pose-m2000.yaml` and the files referenced from it.
 - Batch size: `4 x 8 = 32`
 - For 8 GPUs the required command line parameters are `device='0,1,2,3,4,5,6,7'`
`training_config.batch=32`
- **Training Guide:** `'doc/user-guide/YOLO Retraining Guide for Mythic AMP.pdf'`

Accuracy Metric	FP32 Model	ANA8 Quantized Model	ANA8 Retrained Model	ANA8 Retrained Model 100PPM
Keypoint mAP50:95	0.62	0.008	0.599	0.591

The table below shows the performance, power statistics, and tile statistics for the ANA8 model. For more details on performance, power, and tile statistics (e.g. ACE utilization, Weight utilization, and SRAM utilization per tile), see the document Compiler Optimization Report - YOLOv8s Pose Estimation.

Attribute	Value
Mythic NPU Configuration	m2024 (24 ACEs, 6 ACE tiles, 4 MB per ACE tile, 127 mm ² die area) 8 Digital NPU Cores NPU/CPU subsystems options are still under consideration
Latency (High Optimization)	21.92 ms (some variance in results is observed)
Frames per second (High Optimization)	45 fps
Power Consumption (High Optimization)	0.6 W at maximum frame rate (146 fps) 0.4 W at target frame rate (45 fps)
ACE (compute) utilization	71.1%
Weight (flash cell) utilization	17.0M weights (57.0% of capacity)
SRAM utilization	4.8 MB (19.0% of simulated SRAM capacity)

9 YOLOPX

YOLOPX has been modified as follows:

- Removed PSA block.
- Padded some convolutions with less than 8 channels to 8 channels
- Replaced bilinear upsampling with nearest
- Replaced Ghost blocks (containing depthwise convolutions) with non-depthwise bottleneck layers

The model is structured as follows:

- **Input Resolution:** 1280x736x3

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- **Digital Layers:** The final layers of each of 3 detection heads. For each head, this is 3 convolutions in parallel: class (1x192x1x1), reg (4x192x1x1) and obj (1x192x1x1). Note that these digital layers are not included in the power and latency estimates shown below. This functionality will be added when the digital NPU/CPU subsystem is defined.

The training parameters for this model are defined below. Note that the Hyperparameters are defined in an SDK configuration file.

- **GPU:** Four Nvidia A6000
- **Dataset:** BDD100K
- **Hyperparameters:**
 - `$MYTHIC_SDK_ROOT/mythic-model-zoo/configs/yolopx/training/training_config/config_720p.yaml`.
- **Training Guide:** `'doc/user-guide/YOLOPX Retraining Guide for Mythic AMP.pdf'`

Accuracy Metric	FP32 Model	ANA8 Quantized Model	ANA8 Retrained Model	ANA8 Retrained Model 100PPM
Driving Area Segmentation Accuracy	0.979	0.829	0.978	0.977
Driving Area Segmentation IOU	0.887	0.000	0.880	0.875
Driving Area Segmentation mIOU	0.931	0.415	0.927	0.924
Lane Line Segmentation Accuracy	0.834	0.000	0.831	0.801
Lane Line Segmentation IOU	0.275	0.006	0.268	0.249
Lane Line Segmentation mIOU	0.631	0.455	0.627	0.617
Detection Precision	0.047	0.001	0.048	0.043
Detection Recall	0.890	0.016	0.864	0.851
Detection mAP50	0.681	0.000	0.630	0.615

Accuracy Metric	FP32 Model	ANA8 Quantized Model	ANA8 Retrained Model	ANA8 Retrained Model 100PPM
Detection mAP50-95	0.334	0.000	0.290	0.272

The table below shows the performance, power statistics, and tile statistics for the ANA8 model. For more details on performance, power, and tile statistics (e.g. ACE utilization, Weight utilization, and SRAM utilization per tile), see the document Compiler Optimization Report - YOLOPX.

Attribute	Value
Mythic NPU Configuration	m2048 (48 ACEs, 12 ACE tiles, 4 MB per ACE tile, 253 mm ² die area) 1 Digital NPU Core NPU/CPU subsystems options are still under consideration
Latency (High Optimization)	9.41 ms (some variance in results is observed)
Frames per second (High Optimization)	106 fps
Power Consumption (High Optimization)	2.2 W at maximum frame rate (106 fps) 0.7 W at target frame rate (30 fps)
ACE (compute) utilization	72.6%
Weight (flash cell) utilization	49.9M weights (83.4% of capacity)
SRAM utilization	32.1 MB (63.8% of simulated SRAM capacity)

10 BEVFormer

BEVFormer has been modified as follows:

- Use BEVFormer Tiny with 1600x900 camera resolutions instead of the default 800x450 for this model variant
- Exported, converted, and retrained with training structures before switching back to “eval” mode
- Limited constant folding is applied to simple constant/initializer transformations

The model is structured as follows:

- **Input Resolution:** 6x1600x900x3
- **Digital Layers:** Transformer head

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The training parameters for this model are defined below. Note that the Hyperparameters are defined in an SDK configuration file.

- **GPU:** Eight Nvidia A6000
- **Dataset:** nuScenes with CAN bus expansion
- **Hyperparameters:** `$MYTHIC_SDK_ROOT/mythic-model-zoo/configs/bevformer/training_config/tiny_1600x900.yaml`
- **Training Guide:** `doc/user-guides/BEVFormer Retraining Guide.pdf`

Accuracy Metric	FP32 Model	ANA8 Quantized Model	ANA8 Retrained Model	ANA8 Retrained Model 100PPM
mAP	0.2293	0.0000	0.2290	0.2270
NDS	0.2649	0.0275	0.2552	0.2529
Per-class metrics:				
car AP	0.419	0.000	0.421	0.417
truck AP	0.167	0.000	0.179	0.173
bus AP	0.190	0.000	0.190	0.177
trailer AP	0.035	0.000	0.048	0.042
construction_vehicle AP	0.035	0.000	0.026	0.021
pedestrian AP	0.341	0.000	0.332	0.326
motorcycle AP	0.185	0.000	0.186	0.171
bicycle AP	0.197	0.000	0.171	0.156
traffic_cone AP	0.365	0.000	0.373	0.366
barrier AP	0.359	0.000	0.365	0.359

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The table below shows the performance, power statistics, and tile statistics for the ANA8 model. For more details on performance, power, and tile statistics (e.g. ACE utilization, Weight utilization, and SRAM utilization per tile), see the document Compiler Optimization Report - BEVFormer-Tiny.

Attribute	Value
Mythic NPU Configuration	m2072 (72 ACEs, 18 ACE tiles, 4 MB per ACE tile, 380 mm ² die area) 288 Digital NPU Cores NPU/CPU subsystems options are still under consideration
Latency (High Optimization)	31.93 ms (some variance in results is observed)
Frames per second (High Optimization)	31 fps
Power Consumption (High Optimization)	4.5 W at maximum frame rate (31 fps) 4.4 W at target frame rate (30 fps)
ACE (compute) utilization	68.5%
Weight (flash cell) utilization	35.8M weights (39.9% of capacity)
SRAM utilization	5.9 MB (7.9% of simulated SRAM capacity)

11 Pythia

The model is structured as follows:

- **Model Size:** Pythia is a family of models with a wide range in parameter count. We use the 70-million parameter model.
- **Digital Layers:** All layers are digital except for the linear layers, as well as the sum of linear layer outputs at the end of each transformer block.

The training parameters for this model are defined below. Note that the Hyperparameters are defined in an SDK configuration file.

- **GPU:** One Nvidia A6000
- **Dataset:** SlimPajama
- **Hyperparameters:**
 - `$MYTHIC_SDK_ROOT/mythic-model-zoo/configs/pythia/training_config/training_config.yaml`.
- **Training Guide:** `'doc/user-guide/Pythia Retraining Guide.pdf'`

Loss Metric	FP32 Model	ANA8 Quantized Model	ANA8 Retrained Model	ANA8 Retrained Model 100PPM
Validation loss on SlimPajama	3.57	130.	4.78	4.79